

MORITZ GRAUER

Karlsruhe, Germany
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vexury.dev

TECHNICAL SKILLS

Graphics Programming

Vulkan, OpenGL | Ray/Path Tracing | Real-time Rendering | Offline Rendering

Programming

C, C++, Python, Java, C#, CUDA, GLSL

Tools & Engines

Unity, Unreal Engine, Blender, Git, ImGui, GLFW

Specializations

Ray Tracing Optimization, Cache-Efficient Data Structures, Low-Level Performance Analysis

EXPERIENCE

Computer Graphics Researcher

Feb 2022 – Apr 2026

Karlsruhe Institute of Technology (KIT)

- Published research on ray tracing memory optimization (arXiv)
- Developed custom rendering engine with ray/path tracing support
- Optimized GPU rendering pipelines for interactive applications

QA Engineer

Oct 2020 – Sep 2021

Gameforge (Internship + Working Student)

- Tested gameplay functionality and quest systems across multiple MMO titles
- Debugged and verified in-game rewards, progression systems, and content updates
- Collaborated with dev teams on quality standards and bug resolution

Programming Tutor

Oct 2016 – Jul 2019

Karlsruhe Institute of Technology (KIT)

- Mentored CS students in programming and algorithms

PROJECTS

Quack 3 – Game Jam 2026

Playable FPS merging 2D art with 3D gameplay. Built in Unity with custom mechanics and optimized rendering. Available online.

Custom Ray Tracing Engine

Full rendering engine from scratch with ray/path tracing, textures, lighting and interactive controls using GLFW and ImGui.

Computer graphics researcher specializing in real-time rendering, ray tracing optimization, and GPU computing. Seeking to apply research expertise to practical game and graphics development at innovative studios.

EDUCATION

M.Sc. Informatics

2018 – 2021

Karlsruhe Institute of Technology (KIT)

Specialization: Computer Graphics & AI

Exchange Semester

2018 – 2019

Chalmers University, Gothenburg, Sweden

B.Sc. Informatics

2015 – 2018

Karlsruhe Institute of Technology (KIT)

PUBLICATIONS

Minimizing Ray Tracing Memory Traffic through Quantized Structures and Ray Stream Tracing

arXiv, 2023

Novel approach to reducing memory bandwidth in ray tracing through quantized BVH structures

HONORS

Studienstiftung des deutschen Volkes

German Academic Scholarship Foundation (2018 – 2021)

LANGUAGES

German: Native

English: Fluent (written & spoken)

French, Spanish, Swedish: Basic